Class 2: Probability and Events

STA 220
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Overview

This class will cover Module 2

- Module 2: Probability Events

Topics include:

- Probability and Events
- Conditional Probability
- Independence of Events
- Bayes' Rule

Randomness and Probabilities

Randomness and Probability

Randomness and Probability

- Randomness is why probability is central in statistics.
- We cannot know with certainty what will happen in a single experiment.
- If we could, there would be no need to run the experiment.
- However, we can often describe which outcomes are more likely than others.

Randomness and Probability

Example (fair coin):

- We cannot predict whether a single flip is heads or tails.
- But we know both outcomes are equally likely.

Example (biased coin):

- We still cannot predict a single outcome.
- But we know heads is more likely than tails.

Key Idea

•Individual outcomes are uncertain.

•But over many repetitions, outcomes follow a stable and predictable pattern.

Examples of Randomness

- The result of rolling a 6-sided die
- The time that a bus arrives at the bus stop
- The height of the next person who walks into the room
- The sex of the next baby born in a hospital
- The maternal mortality rate in Canada next year

Probability and Outcomes

Probability is how we describe the chance that an outcome occurs in an experiment.

Example: A fair coin has a 50% chance of landing on heads.

Important:
This does *not* mean outcomes must balance in a small number of trials.

•Example: Two coin flips do not guarantee one head and one tail.

Probability Distributions

A probability distribution describes the chances of *all possible outcomes* of an experiment.

It tells us what we should expect **in the long run**, over many repetitions.

Example: For a fair coin, the distribution assigns:

- 50% to heads
- 50% to tails

- Think of it as a model of what happens if we repeat the experiment “forever.”

Events in Probability

When we run an experiment, we may be interested in specific outcomes.

Example: flipping two coins and getting **first a head, then a tail**

More often, we are interested in a **collection of outcomes** rather than a single one.

Example: flipping two coins and getting **at least one head**

Events

An **event** is a subset of all possible outcomes of an experiment.

We usually denote events using **capital letters**.

- Example:
  - $A =$"getting at least one head in two coin flips"

Probability of an Event

We write $P(A)$ to denote the probability that event A occurs.

For probability to be meaningful, it must satisfy:

$$0 \leq P(A) \leq 1$$

Interpretation:

$P(A) = 1$:event A is certain to occur

$P(A) = 0$:event A will never occur

Sample Space and Events

A **sample space** is the set of all possible outcomes of a random experiment. It is usually denoted by S.

Events are subsets of the sample space.

Sample Space and Events

Example: Rolling a 6-sided die

- Sample space:
- $S = \{1,2,3,4,5,6\}$

Probability distribution (fair die):

- $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$

Examples of Events

- **Simple event:**

$$A = \{1\}$$

- **Compound event:**

$$A = \{1, 2, 3\}$$

Key Properties

- $0 \leq P(A) \leq 1$for any event A
- $P(S) = 1$

Sample Space: Die + Coin

Suppose we roll a die and then flip a coin.

The sample space consists of all possible combinations of outcomes.

There are 12 possible outcomes:

coin flip gives heads

coin flip gives 2

$$S = \{(1\text{H}), (1\text{T}), (2H), (2\text{T}), (3H), (3\text{T}), (4H), (4\text{T}), (5H), (5\text{T}), (6H), (6\text{T})\}$$

Die roll gives 1

Die roll gives 1

How do we count outcomes?

When experiments are combined, we can often use multiplication:

Number of outcomes:

$$6\ \textit{(die outcomes)} \times 2\ \textit{(coin outcomes)} = 12$$

Key Idea

The size of the sample space is the number of all possible combinations of outcomes.

Product Rule (Counting Principle)

If one experiment can occur in m ways and a second independent experiment can occur in n ways, then the total number of combined outcomes is:

$$m \times n$$

If ① experiment 1 is n_1 ways
② experiment 2 is n_2 ways
⋮
⑩ experiment 10 is n_{10} ways

$\Rightarrow$ $n_1 \times n_2 \times \cdots \times n_{10}$ total outcomes of a 10 step experiment

Product Rule (Counting Principle)

Suppose my experiments has three parts now:

- Roll a six-sided die → 6 outcomes ⇒ n_1
- Flip a coin ⇒ 2 outcomes ⇒ n_2
- Spin a colour in five-color spinner (equal parts) ⇒ 5 outcomes ⇒ n_3

Let A = event I roll a 6, flip heads and get red ⇒ $\boxed{p(A) = \frac{1}{60}}$

What is the total number of events in S?

$6 \times 2 \times 5 = 60$ total events in S

Complement of an Event

Sometimes we are interested in the probability that an event **does not occur**.

The **complement** of an event A, written A^c ,represents all outcomes where A does not happen.

"A complement"

Key Formula

$$P(A^c) = 1 - P(A)$$

$A =$ "rolling 6"

$A^c =$ "rolling 1, 2, 3, 4, 5"

$p(A) = \frac{1}{6}$

$p(A^c) = \frac{5}{6}$

Check: $p(A^c) = 1 - \frac{1}{6} = 1 - p(A)$

Complement of an Event

Example

Let A= "it rains tomorrow"

Then:

$P(A)$ =probability it rains

$P(A^c)$ =probability it does **not** rain

Suppose:

$$\boxed{P(A) = 0.3}$$

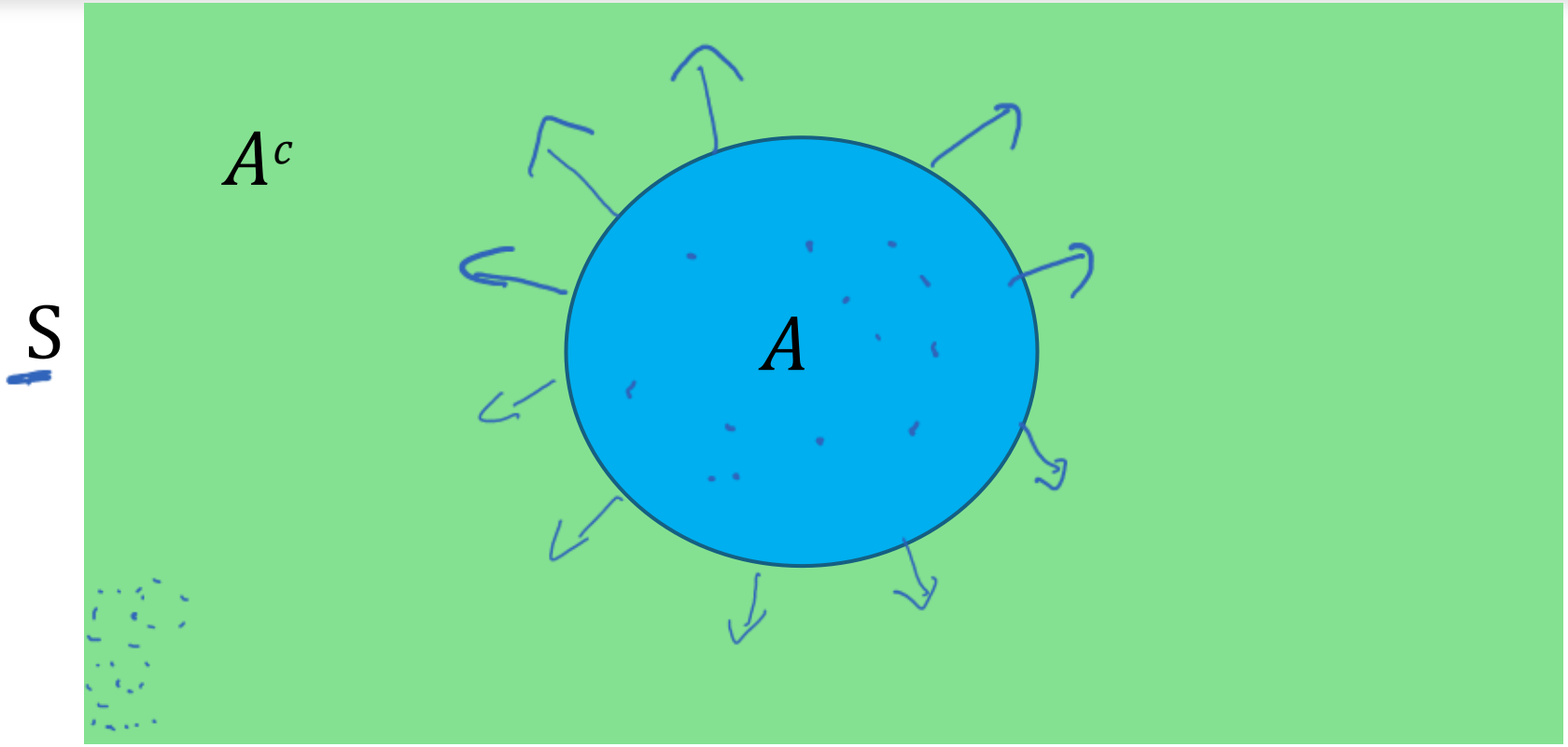
Then:

$$P(A^c) = 1 - 0.3 = 0.7$$

Key Idea

The probability of an event and its complement always add to 1.

Complement of an Event Venn Diagram



Addition Rule

Sometimes we want the probability of **combinations of events**.

Key Rule

$$\boxed{P(A\ or\ B) = P(A) + P(B) - P(A\ and\ B)}$$

What do "or" and "and" mean?

A or B means:

- A happens, or
- B happens, or
- both A and B happen

A and B means:

- both events happen at the same time

Addition Rule

Why do we subtract $P(A\ and\ B)$?

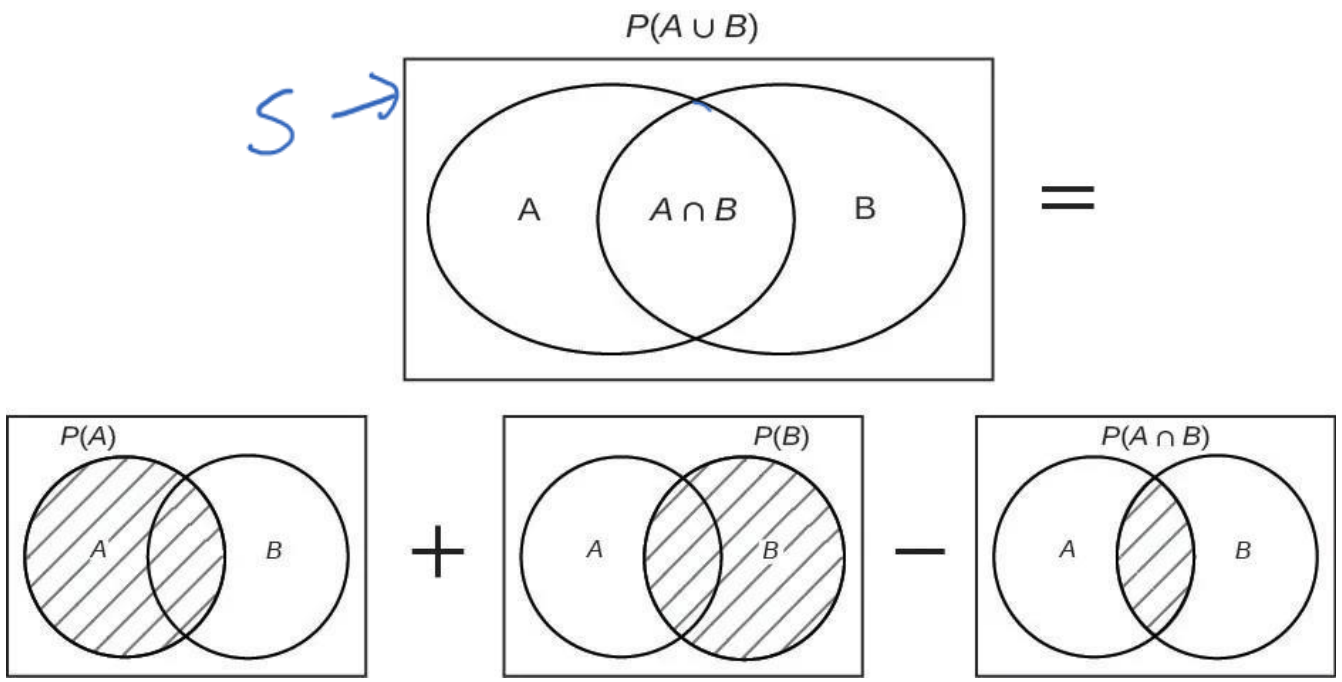
- Because the overlap (both A and B) is counted twice when we add $P(A)$and $P(B)$.

"union"

$A \cup B \Rightarrow A$ or B

$A \cap B \Rightarrow A$ and B

"intersection"



Check Your Understanding!

What happens when I try to use the Addition Rule and events A and B have no outcomes in common?

A. I can't compute P(A or B)

B. I can't compute P(A and B)

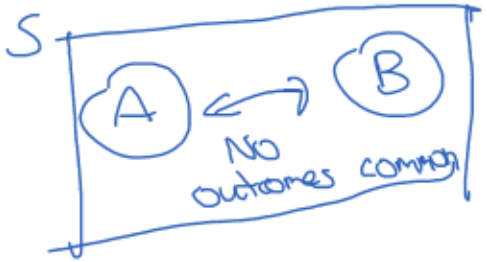
C. I get that $\boxed{\text{P(A or B) = 0}}$

D. I get that P(A and B) = 0

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

addition rule

$P(A \text{ and } B) = 0$



Example: Rolling 2 Dice

When rolling two dice, each die has 6 outcomes.

$S = \{(1,1), (1,2), (1,3), \ldots, (6,6)\}$

- So the total number of possible outcomes is:

$$\underset{1^{st}\text{ die}}{6} \times \underset{2^{nd}\text{ die}}{6} = 36 \rightarrow \text{total outcomes}$$

Possible Sums

- Instead of looking at ordered pairs, we often look at the **sum** of the two dice.

The possible sums are:

$$S = \{2,3,4,5,6,7,8,9,10,11,12\}$$

Example: Rolling 2 Dice

Sum of 2 Dice Chart

1st die

2nd die

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Key Idea

- Different outcomes can produce the same sum.
- Example:

  $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ all give sum = 7

Sum	2	3	4	5	6	7	8	9	10	11	12
# of ways to get it	1	2	3	4	5	6	5	4	3	2	1
Prob	$1/36$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$5/36$	$\frac{4}{36}$	$3/36$	$2/36$	$1/36$

$\Rightarrow$ sum to 1!

Example: Rolling 2 Dice

Example: Sum of Two Dice is NOT 6

When rolling two dice, there are:

$$36 \textit{ equally likely outcomes}$$

The possible sums are:

$$S = \{2, 3, 4, 5, \cancel{6}, 7, 8, 9, 10, 11, 12\}$$

Event of Interest

Let A= "the sum is not 6"

Then:

$$A = \{2,3,4,5,7,8,9,10,11,12\}$$

NOT 6

$$P(\{2, 3, 4, 5, 7, 8, 9, 10, 11, 12\}) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{6}{36} + \cdots +$$

Example: Rolling 2 Dice

Using the Complement Rule

It is easier to use the complement:

Event A^c" =the sum is 6"

There are 5 ways to get a sum of 6:

A = "event not sum 6"

$$(1,5), (2,4), (3,3), (4,2), (5,1)$$

So:

$$P(A^c) = \frac{5}{36}$$

sum of 6

Final Answer

$$P(A) = 1 - P(A^c) = 1 - \frac{5}{36} = \frac{31}{36}$$

Key Idea

Sometimes it is easier to compute the complement first and subtract from 1.

Example: Rolling 2 Dice

Find the following probability that the sum of the 2 dice is at least 4.

$B =$ "sum of 2 dice is at least 4"

$B = \{4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$B^c = \{2, 3\}$

$P(B^c) = P(2) + P(3) = \frac{1}{36} + \frac{2}{36} = \frac{3}{36}$

$P(B) = 1 - P(B^c) = 1 - \frac{3}{36} = \frac{33}{36}$

complement rule!

Example: Rolling 2 Dice

$p(A \text{ and } B) \neq p(A)p(B)$ for any event A and B!

Example: Combining Events (Two Dice)

Suppose we want:
(A) sum is **not 6**
(B) sum is **at least 4**

$\Rightarrow A = \{\cancel{2}, 3, 4, 5, 7, 8, 9, 10, 11, 12\} \Rightarrow p(A) = \frac{31}{36}$

$\Rightarrow B = \{4, 5, \cancel{6}, 7, 8, 9, 10, 11, 12\} \Rightarrow p(B) = 33/36$

We want:

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Addition rule

$A \text{ and } B = \{4, 5, 7, 8, 9, 10, 11, 12\}$

$(A \text{ and } B)^c = \{2, 3, 6\}$

$$p(\underline{A \text{ and } B}) = 1 - p((A \text{ and } B)^c) = 1 - p(\{2,3,6\})$$

$$= 1 - \frac{8}{36}$$

$$= \frac{28}{36}$$

$$p(\underline{\{2,3,6\}}) = \frac{1}{36} + \frac{2}{36} + \frac{5}{36} = \frac{8}{36}$$

$$\Rightarrow p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B)$$

$$= \frac{31}{36} + \frac{33}{36} - \frac{28}{36}$$

$$= \frac{36}{36} = \underline{1}$$

Conditional Probability

Conditional Probability

Conditional probability is closely related to contingency tables and relationships between two categorical variables.

Last week, when we looked at **conditional distributions**, we focused on one group at a time by restricting our attention.

This means we "ignored" part of the table to focus on a specific subgroup.

	Coffee-No	Coffee-Yes	Total-Tea
Tea-No	44	29	73
Tea-Yes	40	45	85
Total-Coffee	84	74	158

tea drinkers

Marginal distrib of coffee drinkers

Conditional Probability

Example (from contingency tables)

- Among **non-coffee drinkers**, what proportion drink tea?
- Here we are only looking at:
  - the subgroup of non-coffee drinkers
  - and then studying outcomes *within that group*

Key Idea

- Conditional probability means we are finding probabilities **given that something is already known**.

	Coffee-No	Coffee-Yes	Total-Tea
Tea-No	44	29	73
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Total-Coffee	84	74	158

$\frac{40}{84}$

Conditional Probability

Example (from contingency tables)

- Among **non-coffee drinkers**, what proportion drink tea? $= \frac{40}{84}$
- Here we are only looking at:
  - the subgroup of non-coffee drinkers
  - and then studying outcomes *within that group*

Key Idea

- Conditional probability means we are finding probabilities **given that something is already known**.

	Coffee-No	Coffee-Yes	Total-Tea
Tea-No	44	29	73
Tea-Yes	40	45	85
Total-Coffee	84	74	158

Total # of Non-coffee drinkers

Conditional Probability

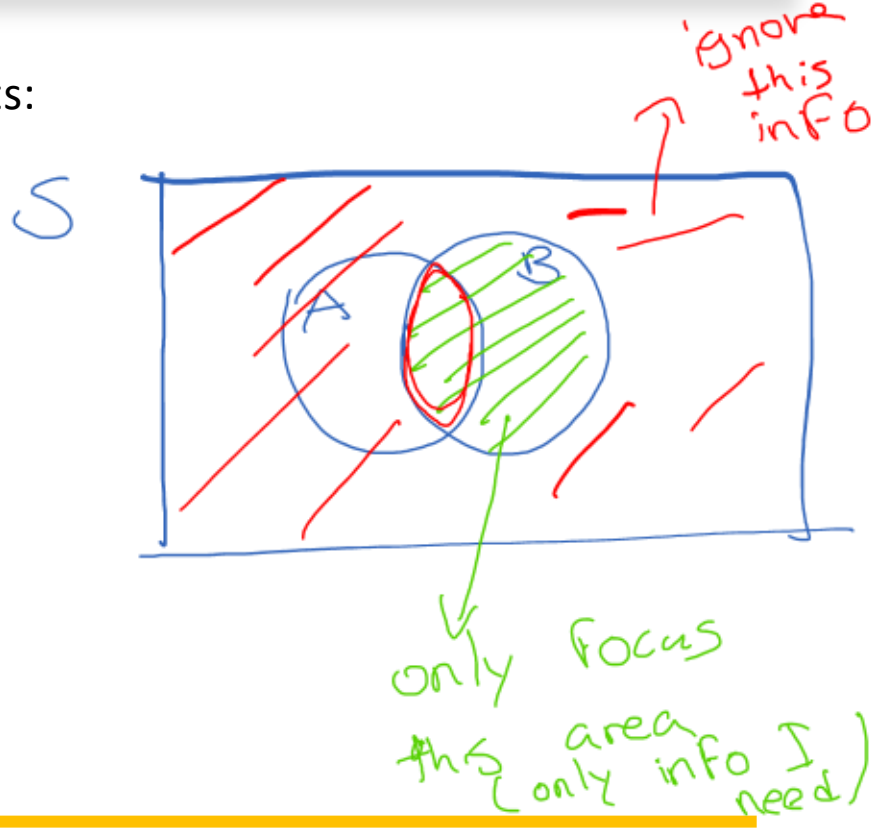
When we compute a conditional probability, we are working with two events:
A: the event we are interested in

- e.g. "tea drinkers"

B: the event that has already occurred (we condition on it)

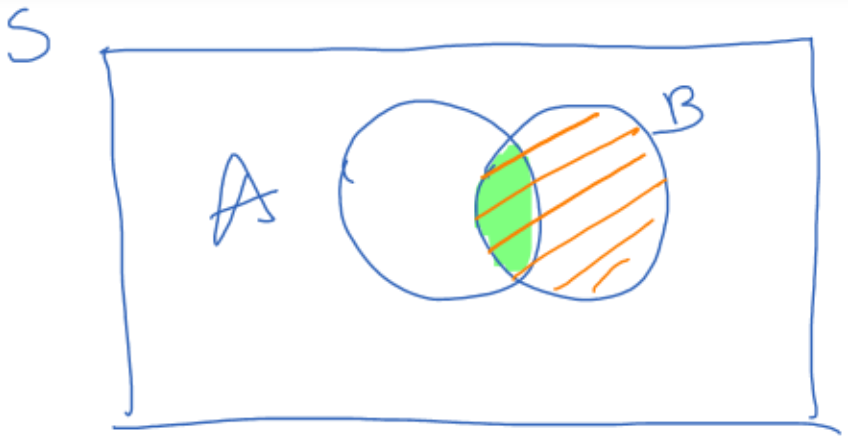
- e.g. "non-coffee drinkers"

Key Idea
Since we know B has happened, we ignore everything outside of B.
We only consider outcomes that lie within B.



Conditional Probability

Venn Diagram Intuition
If we are inside the circle B, we ask:
What is the probability of also being in A?
Think of B as the **new sample space**.



Conditioning on B
We restrict our attention to outcomes in B.
This means removing all outcomes not contained in B.

Key Takeaway
Conditional probability = probability of A **within the reduced sample space** B

Conditional Probability Notation

Definition of Conditional Probability
The conditional probability of A given B is written as:

$$P(A \mid B)$$

"probability of A given B"

Key Idea
When we condition on B, we restrict attention to outcomes where B has occurred.
So we are asking:

"Out of all outcomes in B, what fraction are also in A?"

Conditional Probability Notation

Two-Step Intuition
First, consider outcomes in both A and B: $A \cap B$
Then compare them to all outcomes in B

General Form

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Key Takeaway
Conditional probability is:
probability of the overlap $A \cap B$, **relative to** B

We are no longer using the full sample space — only B

Conditional Probability Example: Coin Flipping

Suppose we flip a coin 3 times.
The sample space has 8 equally likely outcomes:

$$S = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{THT}), (\mathrm{HTT}), (\mathrm{TTH}), (\mathrm{TTT})\}$$

Event of Interest

Let A= "two or more heads"
Then:

$$A = \{(\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{HHH})\}$$

So there are 4 outcomes in A.

Conditional Probability
Example: Coin Flipping

Probability

$$\boxed{P(A) = \frac{4}{8} = \frac{1}{2}}$$

Key Idea

Probability = (number of favourable outcomes) ÷ (total number of outcomes)

All outcomes are equally likely in this experiment.

Conditional Probability
Example: Coin Flipping

Flip a coin 3 times. The sample space is:

S= $\{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{THT}), (\mathrm{HTT}), (\mathrm{TTH}), (\mathrm{TTT})\}$

Let:

- A= "two or more heads"
- B= "heads on the first toss"

Conditional Probability
Example: Coin Flipping

Conditioning on B

If we know the first toss was heads, we remove all outcomes that do not begin with H.

Our new sample space becomes:

$$\boxed{B = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{HTT})\}}$$

New sample space

$= \{(H,H,T), (H,T,H), (T,H,H), (H,H,H)\}$

Which outcomes are also in A?

Outcomes with at least two heads:

$$A \cap B = \{(\mathrm{HHH})\}, (\mathrm{HHT}), (\mathrm{HTH})\}$$

There are:

3 favourable outcomes

4 total outcomes in the new sample space

$P(A|B) = \frac{3}{4}$

Conditional Probability Example: Coin Flipping

Conditional Probability

$$P(A \mid B) = \frac{3}{4}$$

Key Idea

Once we condition on B, we no longer use all 8 outcomes.
We only work within the reduced sample space defined by B.

Conditional Probability
Example: Back to Coin Flipping

Using the Conditional Probability Formula

The sample space is:

S= $\{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{THT}), (\mathrm{HTT}), (\mathrm{TTH}), (\mathrm{TTT})\}$

Let:

A= "two or more heads"

B= "heads on the first toss"

$$p(A|B) = \frac{p(A \text{ and } B)}{p(B)}$$

Step 1: Find the required probabilities

Outcomes in both A and B:

$$A \cap B = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH})\}$$

So:

$$P(A \cap B) = \frac{3}{8}$$

Outcomes in B:

$$B = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{HTT})\}$$

So:

$$P(B) = \frac{4}{8}$$

Conditional Probability
Example: Back to Coin Flipping

Step 2: Apply the formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{3/8}{4/8} = \frac{3}{4}$$

Key Idea

This gives the same result as restricting the sample space to B then finding how many remaining outcomes are in A

Conditional Probability
Example: Back to Coin Flipping

Step 2: Apply the formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{3/8}{4/8} = \frac{3}{4}$$

Key Idea

This gives the same result as restricting the sample space to B then finding how many remaining outcomes are in A

Conditional Probability
Example: Back to Coin Flipping

Conditional Probability Using a Contingency Table

We can also analyze this problem using a contingency table.

Let:

the **columns** represent whether outcomes are in A or not in A

the **rows** represent whether outcomes are in B or not in B

$A = \{(H,H,T), (H,T,H), (T,H,H), (H,H,H)\}$ 3

$B = \{(H,T,T), (H,T,H), (H,H,T), (H,H,H)\}$

$A^c = \{(H,T,T), (T,T,T), (T,H,T), (T,T,H)\}$

	A=2 or more heads	Not A=1 or fewer heads	Row Total
B= head on first toss	3/8 $P(A \cap B)$	1/8	4/8 $= P(B)$
Not B=tail on first toss	3/8	1/8	4/8
Column Total	6/8	2/8	8/8=1

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{3/8}{4/8} = \frac{3}{8} \times \frac{8}{4} = \frac{3}{4}$$

Conditional Probability
Example: Back to Coin Flipping

Conditional Probability Using a Contingency Table

We can also analyze this problem using a contingency table.

Let:

the **columns** represent whether outcomes are in A or not in A

the **rows** represent whether outcomes are in B or not in B

	A=2 or more heads	Not A=1 or fewer heads	Row Total
B= head on first toss			
Not B=tail on first toss			
Column Total			8/8=1

Check Your Understanding!

TRUE or FALSE

If events A and B are disjoint (have no outcomes in common), then

$$\boxed{P(A \mid B) = P(A)}$$

$P(A \text{ and } B) = 0$

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{0}{P(B)} = \boxed{0 = P(A)}$$

$\boxed{\text{True only if } P(A) = 0}$

Independent vs. Mutually Exclusive Events

Independence of Events

Independence of Events
Two events are called **independent** if knowing that one event occurred does not change the probability that the other event occurs.

Example: Roll a Die and Flip a Coin
Let:

- A= "roll a 3"
- B= "land on heads"

A and B are independent

Question
Does knowing the coin landed on heads give us any additional information about whether the die roll is a 3?
The answer is **no**.
The coin flip does not affect the die roll, so the events are independent.

Key Idea
Independence is about information.
If knowing one event occurred does **not** change the probability of the other event, then the events are independent.

Independence of Events Definition

We can check whether two events A and B are independent in two equivalent ways:

1. Using Conditional Probability

Events A and B are independent if:

$$P(A \mid B) = P(A)$$

This means:

Knowing that B happened does **not change** the probability of A

Similarly:

$$P(B \mid A) = P(B)$$

Knowing A happened does **not change** the probability of B

2. Using the Multiplication Rule

Events A and B are independent if:

$$P(A \cap B) = P(A)\,P(B)$$

Independence of Events: Connection Between the Two Definitions

$$p(A \text{ and } B) = p(A)\,p(B)$$

if A and B are indep

$$p(A|B) = \frac{p(A \text{ and } B)}{p(B)} = \frac{p(A)\,\cancel{p(B)}}{\cancel{p(B)}} = p(A)$$

Independence of Events: Example (Rolling Dice/Coin Toss)

Consider:

$A = \{roll\ a\ 3\}$ $= \{(H,3), (T,3)\}$

$B = \{flip\ a\ head\}$ $= \{(H,1), (H,2), (H,3), (H,4), (H,5), (H,6)\}$

Sample Space:

S={(H,1), (H,2), (H,3), (H,4), (H,5), (H,6), (T,1), (T,2), (T,3), (T,4), (T,5), (T,6)}

Is P(A) = P(A|B) ? yes!

Are A and B indep?

$p(A|B) = \frac{p(A \text{ and } B)}{p(B)} = \frac{1/12}{6/12} = \frac{1}{12} \cdot \frac{12}{6} = \frac{1}{6} = p(A)$ ✓

	A=roll a 3	Not A= did not roll a 3	Probability
B=flip a head	$\boxed{1/12}$	5/12	6/12
Not B=flip a tail	1/12	5/12	6/12
Probability	2/12 = 1/6	10/12	12/12 = 1

$p(A)$

Mutually Exclusive Events

Another important idea is **mutually exclusive events**.

A and B are mutually exclusive

Two events are mutually exclusive if:
They have **no outcomes in common**

This is also called **disjoint events**.

Key Idea
The difference between **independence** and **mutual exclusivity** lies in the overlap:

$$P(A \cap B)$$

Mutually Exclusive vs Independent Events

1. Independent Events

- Events **can occur together**
- There may be overlap between A and B
- The overlap satisfies:

$$\boxed{P(A \cap B) = P(A)P(B)}$$

Common outcomes are possible

2. Mutually Exclusive Events

Events have **no overlap**

They cannot happen at the same time

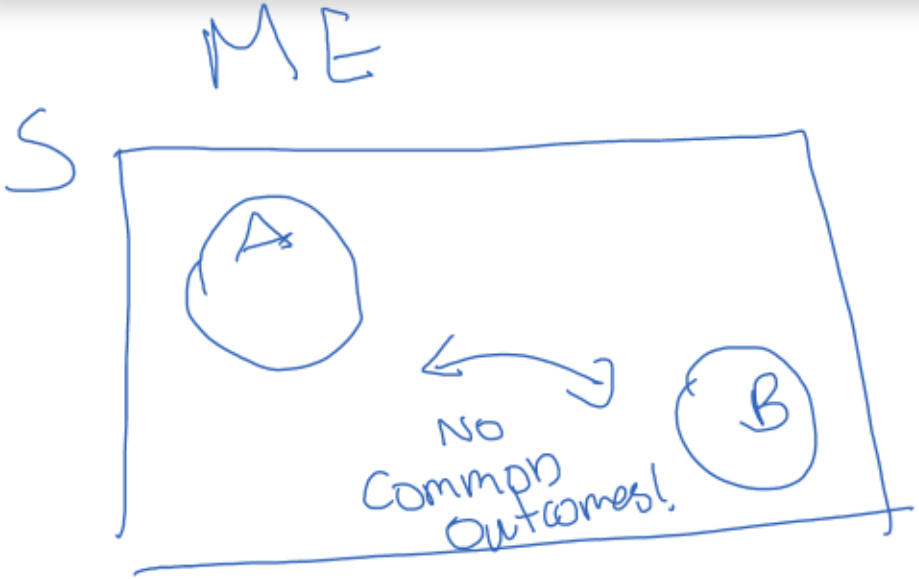
$$P(A \cap B) = 0$$

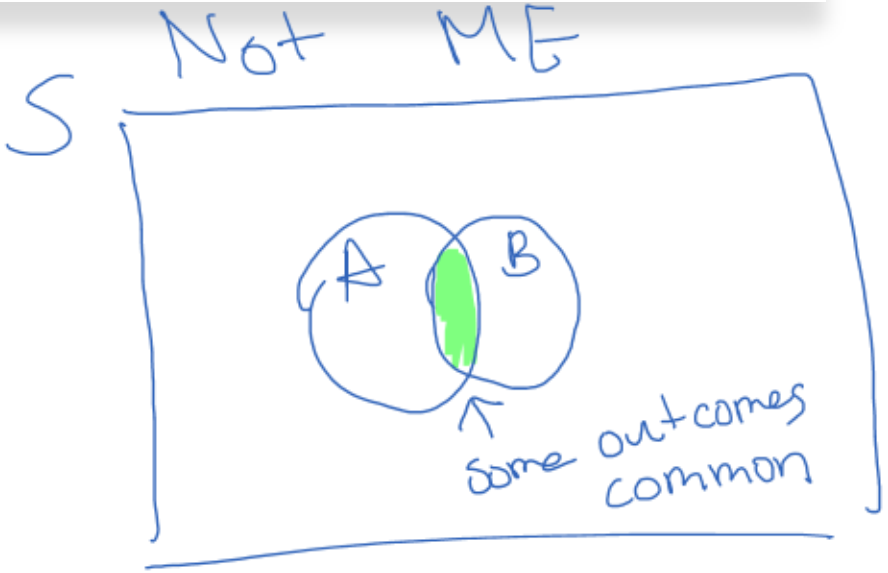
No common outcomes at all

Mutually Exclusive vs Independent Events

Type	**Overlap ($(\mathbf{A} \cap \mathbf{B})$)**	**Interpretation**
Independent	Possible	Occur together, but unrelated
Mutually exclusive	Empty set	Cannot occur together

Mutually Exclusive Events Venn Diagram





Check Your Understanding!

Question: Can mutually exclusive events ever be independent?

$0 = p(A \text{ and } B) = p(A)p(B)$

$\Rightarrow$ Only if $P(A) = 0$ or $P(B) = 0$ or both have 0 prob.

Example: Blood Types

According to Canadian Blood Services:

Type O: 46% **Type A:** 42% **Type B:** 9% **Type AB:** remaining 3%

For one randomly selected person:

Let

$$A = \{\textit{person has Type A blood}\}$$
$$B = \{\textit{person has Type B blood}\}$$

Are A and B mutually exclusive, independent, or neither?

Example: Blood Types

We now consider **two different individuals**.

Let:

$$A^* = \{\textit{first person has Type A blood}\}$$
$$B^* = \{\textit{second person has Type B blood}\}$$

Question

Are A^* and B^* mutually exclusive , independent or neither?

Example: Blood Types

$p(A \text{ and } B) = p(A)\,p(B)$ ONLY!!! if A and B are indep!

According to Canadian Blood Services:

Type O: 46% **Type A:** 42% **Type B:** 9% **Type AB:** remaining 3%

$p(AB) = 0.03 \Rightarrow p(AB^c) = 1 - 0.03$

Among 4 potential donors:

a) What is the probability that they are all Type O?
b) What is the probability no one is Type AB?

a) $p(\text{all 4 have Type O})$
$= p(\{1^{st} \text{ individ type O}\} \text{ and } \{2^{nd} \text{ type O}\} \text{ and } \{3^{rd} \text{ type O}\} \text{ and } \{4^{th} \text{ type O}\})$
$= p(1^{st} \text{ individ type O})\, p(2^{nd} \text{ type O})\, p(3^{rd} \text{ type O})\, p(4^{th} \text{ type O})$
$= 0.46 \times 0.46 \times 0.46 \times 0.46$
$= 0.04477$

b) $p(1^{st} \text{ not AB}, 2^{nd} \text{ not AB}, 3^{rd} \text{ not AB}, 4^{th} \text{ not AB})$
$= p(AB^c)\, p(AB^c)\, p(AB^c)\, p(AB^c)$
$= (1-0.03)(1-0.03)(1-0.03)(1-0.03)$
$= 0.97(0.97)(0.97)(0.97) = 0.8852$

Check your understanding!

Info 1: The probability of event A occurring is 0.3 $\Rightarrow p(A) = 0.3$

Info 2: The probability of event C not occurring is 0.78 $\Rightarrow p(C^c) = 0.78 \Rightarrow p(C) = 0.22$

Info 3: The probability of event A and C occurring is 0.2 $\Rightarrow p(A \text{ and } C) = 0.2$

a) What is the probability that event A occurs if we know C occurred?

$$p(A \mid C) = \frac{p(A \text{ and } C)}{p(C)} = \frac{0.2}{0.22}$$

b) What is the probability that event C occurs if we know A occurred?

c) Are events A and C independent?

d) Ignore Info 2. If they were independent, what would be the value of P(C)?

b) $p(C|A) = \dfrac{p(A \text{ and } C)}{p(A)} = \dfrac{0.2}{0.3} = \underline{2/3}$

c) $p(C|A) = 2/3$

$p(C) = 0.22$

$p(C|A) \neq p(C)$

$\Rightarrow$ A and C are <u>NOT</u> independent!

d) $p(C|A) = p(C) = 2/3$ (ignoring <u>info 2</u>)

Bayes' Rule

Bayes' Rule

Bayes' Rule is a useful result that lets us:

- use conditional probabilities we already know
- and **reverse them** to find a probability of interest

It is a **direct extension of the definition of conditional probability**.

It allows us to "flip" conditioning from

$$P(A \mid B)$$

to

$$P(B \mid A)$$

In many real problems:

$P(A \mid B)$

is easy to think about or given but we actually need

$P(B \mid A)$

Bayes' Rule connects the two.

Tree diagrams

Tree diagrams are a useful tool for working with **conditional probabilities**.

They help us:
- organize outcomes step-by-step
- keep track of all possible combinations of events

How to Use a Tree Diagram
Multiply along branches (left to right)
Move across the tree
Multiply probabilities along each path

$$P(path) = multiply\ along\ branches$$

Add outcomes in the same column (vertical grouping)
Combine probabilities of different paths leading to the same event
Add probabilities within the same category

$$P(event) = add\ relevant\ paths$$

Bayes' Rule and Tree Diagrams Example

We look at the probability structure for two children.

First Birth

Probability first child is **Female (F)**:

$$P(F_1) = 0.5$$

Probability first child is **Male (M)**:

$$P(M_1) = 0.5$$

Second Birth (Conditional)

If the first child is **Female**:

$$P(F_2 \mid F_1) = \frac{1}{3}$$

If the first child is **Male**:

$$P(F_2 \mid M_1) = \frac{2}{5}$$

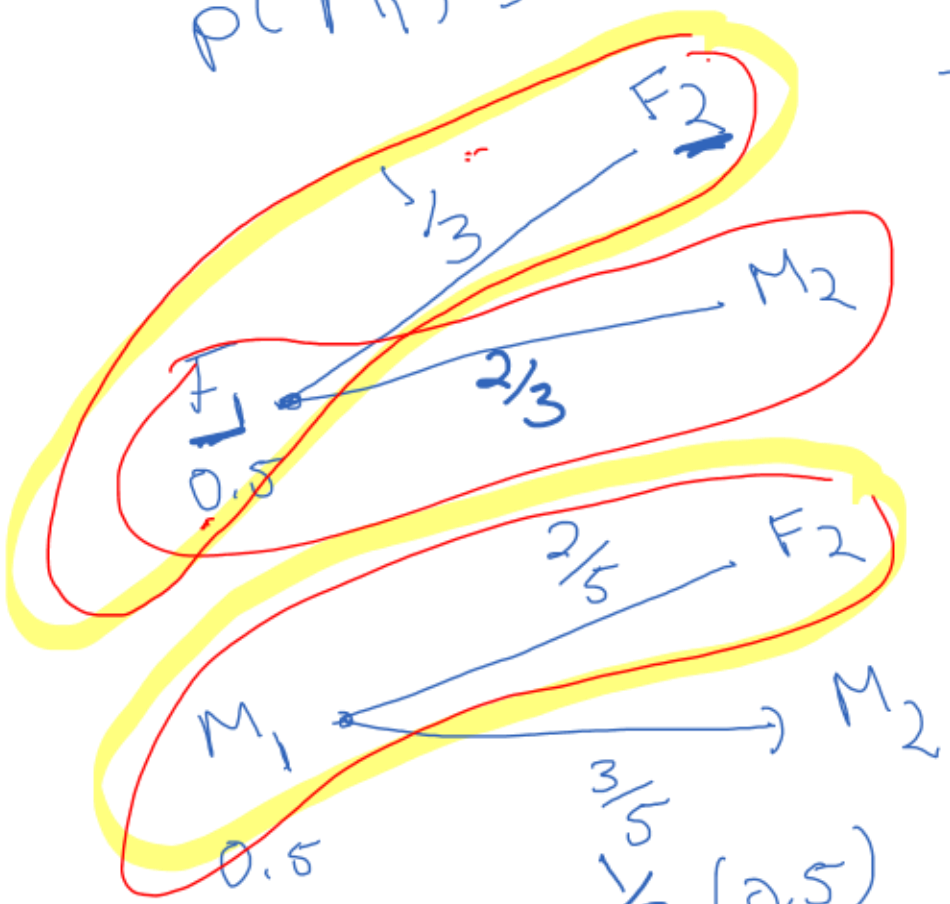
Suppose we want to know the probability of a female first child given that we had a female second child: $P(F1|F2)$.

$p(F_1) = 0.5$

$p(M_1) = 0.5$

$p(F_2 \mid F_1) = 1/3$

$p(F_2 \mid M_1) = 2/5$



branches in 1 root event must all add up to 1

$\Rightarrow p(M_2 \mid M_1) = 3/5$

$p(M_2 \mid F_1) = 2/3$

$\Rightarrow p(F_1 \text{ and } F_2) = 1/3\,(0.5)$

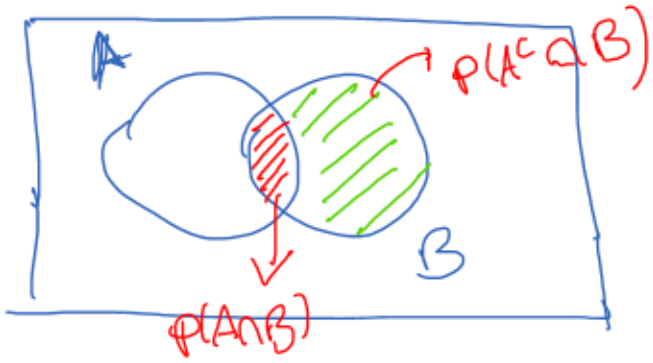
$\Rightarrow p(F_1 \text{ and } M_2) = 2/3\,(0.5)$

$\Rightarrow p(M_1 \text{ and } F_2) = 2/5\,(0.5)$

$\Rightarrow p(M_2 \text{ and } M_1) = 3/5\,(0.5)$

$$p(F_1 \mid F_2) = \frac{1/3\,(0.5)}{1/3\,(0.5) + 2/5\,(0.5)}$$

Interpreting Baye's Rule



From the tree diagram, we can rewrite the components of Bayes' Rule.

Numerator

$$P(A \cap B)$$

This is the probability that **both** A **and** B **occur**.

Denominator

$$P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$$

This can be rewritten using the law of total probability:

$$= P(A \cap B) + P(A^c \cap B)$$
$$= P(B)$$

So Bayes' Rule becomes:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Final Conclusion

Bayes' Rule is simply:
another way of writing the definition of conditional probability

It follows directly from:
- the definition of conditional probability
- and the law of total probability

Bayes Rule

$$p(A|B) = \frac{p(A|B)\,p(B)}{p(B|A)\,p(A) + p(B|A^c)\,p(A^c)}$$

Example: Genetic Testing for Thrombosis

A genetic test is used to determine whether a person has a predisposition for thrombosis, a condition involving blood clots that can block blood flow. It is known that approximately 3% of the population has this predisposition.

The test is highly accurate: if a person has the predisposition, the test correctly returns a positive result 99% of the time. If a person does not have the predisposition, the test correctly returns a negative result 98% of the time.

The question of interest is: if a person tests positive, <u>what is the probability that they actually have the predisposition?</u>

T = "event person actually thrombosis"

$+$ = "person test positive for thrombosis"

$p(T) = 0.03 \Rightarrow p(T^c) = 1 - 0.03 = 0.97$

Question:

$p(T \mid +) = ?$

$p(+ \mid T) = 0.99$

$p(- \mid T^c) = 0.98$

$p(+ \mid T^c) = 1 - p(- \mid T^c) = 1 - 0.98 = 0.02$

$$p(T \mid +) = \frac{p(+ \mid T)\, p(T)}{p(+ \mid T)\, p(T) + p(+ \mid T^c)\, p(T^c)}$$

$$= \frac{0.99\,(0.03)}{0.99\,(0.03) + 0.02\,(0.97)} =$$

(*) Double check using Tree diagram

Lecture Summary

- In this lecture, we covered both parts of Module 2: Probability Events.
- **Practice Problems and Solutions are available on Quercus**
  - See Class 2 Course Materials on Quercus)

Looking Ahead

- **Quiz 2 (and bonus AI usage survey) is due on May 14 at 11:59 PM** (opens on May 11 at 12:00 AM)

- Class 3 (Tuesday) we will cover Random Variables